

Appendix 4: Formulae sheet for examinations

Mensuration
Surface area of sphere = $4\pi r^2$ Curved surface area of cone = $\pi r \times \text{slant height}$ Volume of sphere = $\frac{4}{3}\pi r^3$
Series
Arithmetic series Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$ Geometric series Sum to n terms, $S_n = \frac{a(1-r^n)}{(1-r)}$ Sum to infinity, $S_\infty = \frac{a}{1-r} \quad r < 1$ Binomial series $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots \quad \text{for } x < 1, n \in \mathbb{Q}$
Calculus
Quotient rule (differentiation) $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
Trigonometry
Cosine rule In triangle ABC: $a^2 = b^2 + c^2 - 2bc \cos A$ $\tan \theta = \frac{\sin \theta}{\cos \theta}$ $\sin(A+B) = \sin A \cos B + \cos A \sin B$ $\sin(A-B) = \sin A \cos B - \cos A \sin B$ $\cos(A+B) = \cos A \cos B - \sin A \sin B$ $\cos(A-B) = \cos A \cos B + \sin A \sin B$ $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
Logarithms
$\log_a x = \frac{\log_b x}{\log_b a}$